

World Design & Asset Reference

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World Design & Asset Reference — Humans vs Rebels

This document describes how the game works — art style, campaign layer, tactical battles, resources, bodies, buildings, defenses, fleets, ships, and research — as a reference for anyone (human or AI) creating art, audio, or UI assets for it. This is the full target design, not a description of any one codebase's current feature-completeness. See Section 16 for an honest status check on what's actually built and playable today versus what's described here.

Where a number, cost, or stat below is stated as an example rather than a locked balance value, it's marked as such — this document is a **rules and structure reference**, not a final balance spreadsheet.

1. The pitch

Two factions — **Humans** and **Rebels** — fight over a galaxy on two connected layers:

1. A **turn-based strategic campaign** on a square-grid galaxy map: explore, claim territory, build, research, and move fleets.
2. A **tactical battle layer**: when opposing armies meet, play shifts to a separate battle map for that engagement, in the spirit of *Total War: Three Kingdoms* — armies deploy, then fight it out with individually commanded units on a battlefield that reflects the environment they met in.

There is no diplomacy and no ship-blueprint editor. Ship classes are fixed per faction. Everything else — resources, territory, tech — is deliberately kept simple compared to a traditional 4X: two resources, one way to take territory (conquest through combat), and a small per-faction tech tree.

The whole game is presented as if it were **drawn by hand in a notebook** — see Section 2 for the full art direction. That style is treated as a core part of the design, not a skin applied afterward.

2. Art direction: notebook doodle style

The entire game is styled as **scribbled hand-drawn doodles on notebook paper** — this is the visual identity that should inform every asset, not just a theme for the map.

The world is a notebook

- The galaxy map **is** a page of grid/graph-paper (a math notebook), and the battlefield in a tactical battle is drawn the same way — same square-grid paper look, so the two layers feel like one consistent sketchbook rather than two different art styles.

- The background of both the strategic map and the battle map has hand-drawn stars, and every celestial body (planets, moons, asteroids, stars) is rendered as a doodle too — nothing photographic or rendered, everything looks like it was drawn with a pen on the page.

Strength is in motion and impact, not polish

- The core visual appeal is meant to come from **animation and effects** that look like a **flip book** — hand-drawn, frame-by-frame motion with a raw, slightly-off, charming quality, not smooth vector/3D animation.
- **Shadows** are strong-contrast and hand-drawn (bold hatching/line work), not soft gradients.
- **Hit effects** use **comic-book-style action lines** — radiating speed lines and impact bursts drawn the way a comic panel would show a hit, not a realistic spark/particle effect.
- **Ship destruction** has a specific gag: when a ship is destroyed, tiny hand-drawn **stick figures** are sometimes ejected from it (its "crew" flying free) before it's gone — a deliberate moment of character and humor in an otherwise chaotic explosion, consistent with the notebook- doodle tone throughout.

Faction color and material identity

- **Humans** — metal and **blue** tones; sophisticated, strong, put-together designs. (This already matches the in-project Human government color, `#3447c7`.)
- **Rebels** — a rusty, patched-up scrapyard look; earthy tones (Mars red, rust orange, etc.) rather than clean metal. Ships should read as repurposed/salvaged rather than manufactured. (This already matches the in-project Rebel government color, `#c73434`, at the red end of that earthy palette.)
- Combined with the propulsion/weapons split in Section 11 (Humans = fire/combustion, Rebels = ion/laser), color and material alone should be enough to tell the two factions apart even in silhouette.

3. The galaxy map

The galaxy is organized in two tiers: a wide map of **areas** (like systems), and each area is itself a local **square grid** you enter and maneuver within.

Areas

- The galaxy is made of discrete **areas**, connected to their neighbors — moving a fleet from one area into an adjacent one is how a player actually travels the galaxy.
- **Undiscovered areas show as grey/blank.** An area is only revealed the moment a fleet actually enters it — at that point everything inside becomes visible all at once (its body/bodies, any fleets present), not gradually.
- Each area usually holds one body, occasionally more: a planet, a moon, or an asteroid belt (Section 6).

- Map size is configurable (small/medium/large); a larger setting adds more areas around the map.
- The two players start on opposite outer edges of the map.

Inside an area

- Once entered, an area is its own **10×10 grid of square fields**, drawn like graph/math-notebook paper (Section 2) — simpler to draw and animate than a hex grid, and it directly reinforces the notebook art direction.
- Bodies sit at the **center** of their area, with a fixed footprint:
 - Planet: **3×3 fields**.
 - Moon: **2×2 fields**.
 - Asteroid belt: **3×3 fields** (same footprint as a planet).
- **Ships are tiny by comparison — a single ship fits inside one field**, and many ships can occupy the same field without visual crowding, which is part of why armies can run up to 50 ships (Section 9).
- Selecting a discovered body shows its output, owner, build slots, and structures.

Movement, engagement, and claiming distance (inside an area)

- Movement speed is spent in **fields per turn**. A tuned baseline: Small ships move **5 fields per turn** (an initial pass used 3, but that felt too slow to cross a typical 10×10 area in a reasonable number of turns).
- **Triggering a battle**: two hostile armies trigger a battle once they close to within **2 empty fields** of each other.
- **Claiming a body**: a fleet must be within **1 field** of the body's edge to use the Claim action — see Section 10 for exactly what claiming does depending on who (if anyone) already owns the body.

4. Turn structure

The strategic campaign layer is **turn-based**. Each turn resolves in this order:

1. Resolve queued construction and ship production.
2. Apply Minerals and Energy income (and subtract Energy upkeep — see Section 5).
3. Advance research.
4. Heal/repair fleets sitting in friendly territory.
5. Reset each unit's movement allowance for the new turn.
6. Players (or the CPU) act: move fleets, explore, claim, build, queue research, form/merge fleets, and initiate engagements.
7. End turn.

A fleet's movement allowance per turn is capped by its **slowest ship's speed stat** — the whole army only moves as fast as its slowest member. Movement allowance is spent, not banked; a unit can't save up unused movement from a previous turn or be issued multiple move commands to exceed its per-turn allowance.

5. Resources

Exactly two resources exist — no Influence, no native resources (Ore/Water/ Gems/etc.), no population or labor as a player-facing stat.

Minerals

- Pays for structures and ships.
- **Finite per body.** Every mineral-producing body (planets, asteroid belts) holds a limited pool; mining it down is possible, unlike Energy. Once a body's pool is exhausted, its Mineral output drops off — this is a real constraint on how long a single claimed body keeps paying off, not an infinite tap.
- Spent in one-off amounts on buildings and ships (not a continuous drain).

Energy

- Pays for research **and** is the upkeep resource for your military and economy: every ship costs Energy per turn to maintain, and abilities/ research/hyperdrive jumps/stargates can also cost Energy to use.
- **Not finite** — a body that can produce Energy can always be harvested for more of it (subject to which buildings/research you've unlocked there). The limiting factor for Energy isn't "running out," it's whether your production stays *ahead* of your total upkeep.
- **Worked example:** if your empire produces 10 Energy per round and you have 5 Small ships, each costing 1 Energy/round in upkeep, your total upkeep is 5 — so your *net* Energy gain that round is $10 - 5 = 5$, not 10. Add more ships (or activate more Energy-costing abilities) than your production can support, and your net goes to zero or negative — Energy is simultaneously your currency, your research fuel, and the hard cap on how large a fleet you can actually sustain.

Starting conditions

- After picking a faction and a map, both players begin with **50 Minerals and 50 Energy**.
- Humans start with **one Small ship** and a home planet named **Eden**.
- Rebels start with their Capitol ship and escorts (Section 11) and no home planet at all (Eden's equivalent is the Capitol itself).

Spending: buildings, ships, and research cost both resources

Queuing a building or ship, or starting a research project, spends **both** Minerals and Energy together — not Minerals alone. **Energy is typically the larger of the two costs.** The example costs given elsewhere in this document (Section 7's building table, etc.) show the Minerals side only; treat those as illustrative, with an Energy cost on top still to be tuned.

UI: resource display

The top-left of the HUD always shows both current totals, plus a small "+" indicator next to each showing the **projected gain for next round** — so a player can see their income trend before it actually lands, not just their current balance.

Both resources are always visible in the HUD.

6. Heavenly bodies

Every discoverable object type can be owned and built on. Each has its own build-slot count and resource profile:

Body	Build slots	Minerals	Energy	Notes
Planet	5 (up to 8 on rare "special" planets)	Finite pool, moderate	Harvestable, moderate	The standard ownable body; occupies a 3×3 block of fields at the center of its area (Section 3). The Human starting planet, Eden, is one of these.
Moon	3–4	Weak	Weak	Found orbiting some planets; occupies a 2×2 block; not worth as much as a planet or asteroid belt
Asteroid belt	2–3	Finite pool, faster/bonus extraction rate than a planet	None (Minerals only)	Mineral-specialized; occupies the same 3×3 footprint as a planet; per-round mining bonus makes it the best pure-Minerals body
Star/sun	Energy-harvester slots only	None	Massive amounts, harvestable	Building a basic harvester works immediately; a specialist stellar harvester (bigger yield) requires research to unlock
Space station/satellite	2	Depends on what's found	Depends on what's found	Not built from scratch by a player directly — found as a derelict and repaired (Section 14), or otherwise claimed. Each one carries one random bonus : extra Energy, extra Research, extra Minerals, a free Spaceport, or a batch of ships

Some systems generate a **planet-and-moon pair** — the moon is a secondary, weaker body in the same system as its planet, not a separate discovery.

A body produces nothing until owned. The only way to gain ownership, neutral or enemy, is conquest through combat (Section 10) — there is no free/instant "settle an empty planet" action.

Selecting a body

Clicking an owned or claimable body shows its build slots as small square buttons directly next to its info panel — not a popup or separate menu.

- An **empty square** shows a "+" — click it to choose what to build there.
- A **filled square** shows that building's icon — click it to destroy it.
- A body with a Spaceport gets an extra ship-shaped button for queuing ship production (ships go into a build queue, they don't occupy a slot).

No terrain, biome map, or population figure is shown for any body — none of that is modeled, so nothing should visually imply it exists.

7. Buildings (economic slots)

Building	Cost (Minerals, example)	Effect	Buildable on
Mineral Mine	2	Extracts from the body's finite Mineral pool	Planet, Moon, Asteroid belt
Energy Harvester	2	Harvests Energy (basic rate)	Any body that can produce Energy
Specialist Asteroid Harvester	research-gated	Bonus-rate Mineral extraction on an asteroid belt	Asteroid belt only, after the relevant research
Specialist Stellar Harvester	research-gated	Large-yield Energy harvesting from a star	Star only, after the relevant research
Spaceport	5	Unlocks ship construction on this body	Any body
Research Lab	5	Enables/funds research	Any body

The "specialist" harvesters are deliberately gated behind research rather than available from turn one — they represent an economic tech investment, not a starting option. Buildings finish a turn after being queued; their cost is deducted immediately on queuing. Costs above are the Minerals portion only — every building also costs Energy, usually more than the Minerals shown (Section 5).

8. Body defenses (separate defense-platform slots)

Every ownable body has a **second, separate set of slots** just for defenses — distinct from the economic building slots in Section 7. These don't produce Minerals or Energy; they exist purely to help defend that body if it's attacked.

Defense structure	Effect
Rocket Battery	Fires on enemy ships during a battle at this body
Laser	Direct-fire defensive weapon
Forcefield	Shields friendly units stationed near it during battle
Disruptor	Periodically (roughly every 10 seconds) jams enemy units in range, preventing them from moving for a few seconds

How defenses factor into a battle

If an enemy attacks a tile where you have a body with defenses (and possibly a fleet also stationed there), the battle is fought as **your fleet plus that body's defenses, together, against the attacker** — the defense platforms aren't a separate fight, they're extra combatants on your side of the same battlefield. Where you place your fleet on the tile/ battlefield relative to those fixed defenses matters, tying into the deployment step described in Section 10.

9. Fleets and armies

- Ships on the same tile can be grouped into an **army** (fleet).
- An army's cap is **50 ships**.
- Armies can be **merged** (combine two friendly armies into one) and move together as a single unit once merged.
- An army's movement speed per turn equals its **slowest ship's** speed stat (Section 4).
- Every individual ship costs Energy per turn to maintain (Section 5) — this is the real limiter on how large a standing military an empire can field, separate from the one-time Mineral cost to build the ships in the first place.

Fleet UI

- The player's own fleet is shown in the **bottom-right** of the HUD as a row of ship slots, one per ship — these can be manually **reordered by strength or defense**, which matters for how the fleet lines up during the battle prep phase (Section 10).
- A separate, dedicated slot in the same panel holds the fleet's assigned **Commander** (Section 14).

10. Conquest and tactical battles

There is one way to take territory: defeat whoever's defending it, then claim it. There is no diplomacy-based annexation.

The campaign is turn-based; battles are not. Only the strategic layer (Section 4) runs in discrete turns. The moment a battle starts, control switches to continuous real-time RTS play for the duration of that fight — there is no "turn" inside a battle, units move and act continuously, the same as any real-time tactics game.

Triggering a battle

Two hostile armies trigger a battle once they close to within **2 empty fields** of each other (Section 3). If the target holds a body with defenses (Section 8), those defenses join the defending side automatically. Both players ready up to begin.

Claiming a body

Once your fleet is within **1 field** of a body's edge (Section 3), you can use the Claim action — what happens depends entirely on who (if anyone) already owns it:

- **Empty/neutral body:** ownership transfers **immediately** — no fighting required at all.
- **Enemy-owned body with an active defense platform:** the defense platform must be destroyed first, in a full tactical battle (this section) — only once it's gone can a claim be made.
- **Enemy-owned body with buildings but no defense platform:** pressing Claim doesn't resolve instantly. Each building on the body has its own defense value by type — **Mineral Mine / Energy Harvester: 5, everything else (Spaceport, Research Lab, etc.): 10**. Your fleet's strength grinds this down over time (proportional to how strong your fleet is), destroying buildings one at a time. Once every building is gone, the body can be claimed exactly as if it had always been empty.

An enemy body can therefore be taken two different ways depending on what's defending it: a straight tactical battle if there's a defense platform, or a strength-vs-buildings grind (no separate battle screen needed) if there isn't.

Prep phase

Before the fight starts, there's a **one-minute deployment phase**:

- Each player places their own ships anywhere on their own side of the battlefield.
- When a player is satisfied with their placement, they press **Ready**.
- Once **both** players have pressed Ready, the battle begins.

The battle itself

- The battle takes place on a **separate battle map**, generated to reflect the environment of the strategic tile the encounter happened in (e.g. a battle near a star looks different from one in open space or near an asteroid field) — assets need to support multiple battlefield backdrops/environments for this reason.
- Like the strategic map, the battlefield is drawn as **square-grid notebook paper** (Section 2/3) for visual consistency. Movement itself stays continuous and real-time, not locked to one ship per cell — ships are tiny relative to a square (Section 3), so many can occupy the same area of the grid at once. Units occupy real physical space and **collide/bounce off each other** rather than freely overlapping or fully stacking.
- **Each unit can be commanded individually** — select one or many ships and give them move/attack orders independently, the way a real-time tactics game works, not by giving one order to an entire army at once.
- The battle ends when **every unit on one side's battlefield force is destroyed**.

After the battle

- Surviving ships return to the strategic map with whatever damage they took.
- Damage is **healed/repaired starting the turn after the battle** while a fleet sits in friendly territory (see also Aegis and Scarab, Section 11, which can heal ships *during* a battle rather than waiting for the next turn).
- **Damage carries over** if the same (not-yet-fully-repaired) army is attacked again by another enemy army before it finishes healing — repairs don't reset or protect a fleet from a second engagement.

11. Ships

There is no in-game ship-design editor; ship classes are fixed per faction, and — importantly — **the two factions do not have a mirrored roster**. Both keep Small/Medium/Large-style baseline hulls (Rebels skip Small — see below), and both get **four research-unlocked specialists** on top, but those specialists cover different roles rather than mirroring each other one-for-one. Several specialists have an in-universe nickname alongside their functional name.

Two different "speed" stats — don't confuse them, throughout this whole section. The "Speed" values below are a separate, real-time movement/ agility rating used only inside a tactical battle. They have nothing to do with the strategic fields-per-turn movement from Section 3 (Small ships = 5 fields/turn on the campaign map, locked) — that stat governs the turn-based campaign layer only.

Human roster

Ship	Tier	Role
Small	Baseline	Cheapest, fastest-to-build hull
Medium	Baseline	Mid-tier hull
Large	Baseline	Heaviest baseline hull
Shroud ("Paladin")	Specialist (research-unlocked)	Defensive support — projects a shield onto nearby allies
Artillery ("Longbow")	Specialist (research-unlocked)	Long-range damage and area control
Trebuchet	Specialist (research-unlocked)	Extreme-range siege — very slow, very powerful rocket artillery
Aegis	Specialist (research-unlocked)	Repair and durability support — the Human combat medic

Rebel roster

Rebels have no Small hull. Their lightest and only "baseline" combat ship is the Medium — already the fastest hull in the game.

Ship	Tier	Role
Medium (<i>becomes "Jumper"/"Phantom" once researched</i>)	Baseline	Fast, cheap, only light/medium hull Rebels have
Drone Ship ("Hive")	Specialist (research-unlocked)	Carrier/swarm ship — durable and slow, but its value is in the drones it releases
Lancer	Specialist (research-unlocked)	Extreme-range laser siege — the Rebel counterpart to the Human Trebuchet
Scarab	Specialist (research-unlocked)	Fast armed repair carrier — the Rebel counterpart to the Human Aegis
Capitol	Flagship	Mobile home base and army carrier

Two of these specialists also spawn a small, weak sub-unit that isn't independently built: Drone Ship releases **Microdrones**, and Scarab releases **Scarab Repair Drones**. Both are covered under their parent ship below.

Human specialists in detail

Shroud ("Paladin") — 12 HP, 2 Attack, Speed 2, Range 3, 92% accuracy, 0.9 shots/sec. Weak on its own; its value is entirely its ability.

- **R-ability — Shield Field:** projects a shield around nearby allies. At its base research level: **5 shield HP, 3-field radius**. Higher research levels raise that to **7/9 shield HP and radius 4/5** (a direct worked example for the Level 1/2/3 research system in Section 13). The shield absorbs damage before hull HP, **stops Rebel Penetrating Lasers outright** (Section 13), and has a **75% chance to intercept an incoming rocket** (see "Rocket interception," below). It reactivates automatically once its cooldown finishes.

Artillery ("Longbow") — 35 HP, 12 Attack, Speed 0.8 (slow), Range 5, 70% accuracy, 0.45 shots/sec. A slow long-range gunship.

- **Passive — Shrapnel:** a successful hit also damages units standing around the target (area splash on a direct hit, not a separate toggle ability).
- Its accuracy was originally set to 55%, but missing roughly every other shot felt too frustrating to play against/with, so it was raised to 70% — worth remembering as a concrete example of "accuracy that's too low isn't fun, even if it's mathematically balanced" when tuning other units.

Trebuchet — 45 HP, 30 Attack, Speed 0.6 (very slow), Range 99 (effectively unlimited), 65% accuracy, 0.25 shots/sec.

- **R-ability — Rocket Barrage:** an extremely powerful rocket attack, **12-second cooldown**. Because Shrouds and drones can intercept its rockets in flight (see below), Trebuchet is at its best kept protected and firing from the back of the formation rather than exposed forward.

Aegis — 30 HP, 0 Attack (does not fight), Speed 2.5.

- Automatically follows and stays near damaged allies.
- Continuously repairs any friendly ship within **radius 2** — roughly **2.4 HP/second** in a representative combat simulation — and grants each nearby ally **+5 maximum HP, once** (not a repeating buff).
- This is the ship that fills the "healer" role in this document's earlier drafts (previously called "Repair Ship"); Aegis replaces that concept with a more fleshed-out kit.

Rebel specialists in detail

Jumper / "Phantom" (the Medium, once researched) — 18 HP, 12 Attack, Speed 2.5, Range 2, 95% accuracy, 0.8 shots/sec.

- **R-ability — Phase Jump:** teleports to a selected point after a short charge (base: **~1 second**; higher research levels reduce this to **0.75s then 0.5s** — another concrete Level 1/2/3 worked example). **8-second cooldown**. Meant for flanking, escaping, or jumping onto vulnerable rear-line ships (Trebuchet, Lancer, Aegis, Scarab) rather than brawling head-on.
- Until Phase Jump is researched, this ship is a plain Medium with no R-ability.

Drone Ship / "Hive" — 50 HP, 8 Attack, Speed 1.5, Range 2, 90% accuracy, 0.55 shots/sec. Durable but slow; most of its value comes from its drones, not its own gun.

- **R-ability — Drone Swarm:** releases **10 Microdrones** once per activation.

- **Microdrone** (spawned, not directly built): 1 HP, 0.2 Attack, Speed 5, Range 1, 85% accuracy, 1.6 shots/sec. Extremely fast and fragile — individually negligible, dangerous in numbers. Inherits the same research/commander bonuses as normal ships, can gain the Penetrating Lasers effect once researched, and each has a **50% chance to intercept a nearby rocket**, taking damage in the process (see "Rocket interception," below).

Lancer — 40 HP, 25 Attack, Speed 0.8 (slow), Range 99 (effectively unlimited), 90% accuracy, 0.3 shots/sec. The Rebel counterpart to the Human Trebuchet: same extreme-range siege role, laser instead of rockets.

- Fires a beam rather than a projectile — nothing to intercept the way Trebuchet's rockets can be, but the beam has its own falloff instead: full damage at range 1–2, then roughly **-10% per additional range step**, down to a **15% minimum**. A Shroud's Shield Field still stops the beam from penetrating through to additional ships behind the first target it hits.

Scarab — 35 HP, 8 Attack, Speed 3.5, Range 2, 88% accuracy, 0.9 shots/sec. A fast healer that can still fight a little — the Rebel counterpart to the Human Aegis, but expressed as a mobile drone swarm instead of a single dedicated medic ship.

- **Automatically deploys 10 Scarab Repair Drones** when a nearby ally takes damage, or after 3 seconds of continuous combat — not a manually-triggered R-ability, it reacts on its own.
- **Scarab Repair Drone** (spawned, not directly built): 3 HP, 0 Attack (does not fight), Speed 4.5. Automatically finds the friendliest non-drone ship with the lowest HP percentage, flies to it, and repairs it. All ten together are what actually deliver Scarab's healing output. Like Microdrones, they can also intercept a nearby rocket, taking damage in the process.
- This ship replaces this document's earlier "Suicide Drone" concept — Scarab fills the healer-counterpart role instead of a kamikaze one.

Rocket interception (a mechanic that ties several units together)

Rocket-firing units (Trebuchet's Rocket Barrage; Artillery's Shrapnel hits don't count, only true rocket projectiles do) can have their rockets shot down before impact:

- A **Shroud's active Shield Field** has a **75% chance** to intercept an incoming rocket.
- A **Microdrone** or **Scarab Repair Drone** each individually has a **50% chance** to intercept a nearby rocket — and takes damage doing so. This is why Trebuchet (and its Rebel counterpart Lancer, though Lancer's beam has no projectile to intercept) plays best from behind a protected formation rather than exposed at the front.

Capitol: the moving carrier

The Capitol is best understood as a **mobile aircraft carrier that is also the Rebel faction's home**:

- It doubles as the Rebels' economic "homeworld" — Rebels have no fixed home planet, this ship *is* their home, with its own build slots and its own per-turn Minerals/Energy income.
- It can carry and move together with its entire attached army — up to the full 50-ship army cap (Section 9) — as a single unit on the strategic map.

- **Jump** (its unique strategic ability, not a battle R-ability): teleports the Capitol and its entire attached army a chosen number of grid squares across the *strategic* map. Cost scales with both army size and jump distance: **1 Energy per ship, per field jumped**. A full 50-ship army jumping 1 field costs 50 Energy; jumping 3 fields with the same army costs 150 Energy. Jump has a **2-turn cooldown**.

Combat stats at a glance

Consolidating the specialist stats above alongside the baseline hulls (baseline numbers are still this document's own earlier estimate, not yet replaced with playtested figures the way the specialists above have been):

Ship	Attack	HP	Accuracy	Rate of Fire	Battle Speed	Notes
Human Small	4	15	88%	1.0/s	3	Cheap skirmisher
Human Medium	8	30	85%	0.7/s	2	All-rounder
Human Large	15	60	90%	0.5/s	1	Slow tank
Human Shroud	2	12	92%	0.9/s	2	Value is Shield Field, not its attack
Human Artillery	12 (+ Shrapnel splash)	35	70%	0.45/s	0.8	Slow, long range; R-ability adds Rocket Barrage
Human Trebuchet	30	45	65%	0.25/s	0.6	Extreme range; needs protection from interception
Human Aegis	0 (non-combatant)	30	—	—	2.5	Value is entirely its healing
Rebel Medium / Jumper	12	18	95%	0.8/s	2.5	Fastest hull in the game; gains Phase Jump once researched
Rebel Drone Ship	8	50	90%	0.55/s	1.5	Value is its Microdrone swarm, not its own gun
Rebel Microdrone (spawned)	0.2	1	85%	1.6/s	5	Fast, fragile, dangerous only in numbers
Rebel Lancer	25	40	90%	0.3/s	0.8	Extreme-range beam with range falloff, not interceptable
Rebel Scarab	8	35	88%	0.9/s	3.5	Fast healer that can still fight a little
Rebel Scarab Repair Drone (spawned)	0 (non-combatant)	3	—	—	4.5	Auto-heals the lowest-HP% ally
Rebel Capitol	25	100	92%	0.9/s	1	Not built to brawl — its value is capacity and Jump

In short: **Humans get shields, artillery, rockets, and steady area repair; Rebels get teleportation, attack swarms, long-range lasers, and mobile repair drones.** These specialists were designed to deepen faction asymmetry beyond the baseline Small/Medium/Large hulls, and — being a later addition than the original core design — still need a real balance pass and finished visual assets, same as everything else marked "not final" in this document.

12. Battle-wide abilities (not tied to a specific ship)

Beyond individual ship R-abilities, the design calls for **empire-level superweapon abilities** usable during a battle, independent of any one ship — unlocked through research (Section 13), not built as a unit:

- **Antimatter Bomb / Thermonuclear Device** — a large-area, high-impact strike usable mid-battle. Given the power level implied, these should be scarce: a sensible starting design is a small number of uses per game (or a very long cooldown) and a steep Minerals/Energy cost, rather than a spamable ability. Exact costs, radius, and damage are unspecified and need a dedicated balance pass before implementation.
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13. Research

Research is spent on faction identity and unlocks — not generic economy scaling. Confirmed unlock categories:

- **Specialist harvesters** (Section 7): unlock bonus-rate Mineral extraction on asteroid belts and large-yield Energy harvesting on stars.
- **Special units and upgrades** (Section 11): Shroud, Artillery, Trebuchet and Aegis for Humans; the Medium → Jumper conversion, Drone Ship, Lancer, and Scarab for Rebels — plus follow-on upgrades like Shroud's shield scaling, Jumper's faster Phase Jump charge, and increased Drone Swarm size/output.
- **Stargates**: build a stargate connecting two chosen points on the strategic map; once built, fleets can travel between the two connected points directly instead of moving tile-by-tile. A stargate connection costs Energy continuously, per round, to stay open/maintained — it's an ongoing strategic investment, not a one-time cost.
- **Battle-wide superweapons** (Section 12): Antimatter Bomb / Thermonuclear Device.

Both factions keep their own small, separate tech list (not a shared tree), each research project taking about one turn to complete with adequate Research Lab support.

Example tech list (grounded in prototype history)

An earlier prototype of this game had a working, playtested tech tree. These are good concrete examples of what a real tech list looks like — adapt names/costs to fit this game's economy rather than copying the exact old numbers verbatim:

Human examples:

Tech	Effect
Advanced Extraction	Bonus Minerals/Energy per building
Reinforced Plating	+25% HP, fleet-wide
Battle AI	+25% accuracy
Advanced Railguns	+25% attack
Shield Tech	Unlocks the Shroud
Artillery Systems	Unlocks Artillery (requires a Spaceport)
Shield Matrix	Every Human ship gets a small personal shield (baseline: 1 shield HP) that absorbs one hit before hull HP is touched, doesn't regenerate once spent, and — notably — also blocks Rebel Penetrating Lasers , the same as a Shroud's Shield Field does

Rebel examples:

Tech	Effect
Warp Drive	+25% thrust/speed
Composite Armor	+25% HP, fleet-wide
Enhanced Lasers	+25% attack
Phase Jump	Unlocks Jumper (the Medium's researched upgrade)
Drone Swarm	Unlocks the Drone Ship
Capital Construction	Cheaper Capitol-class construction
Penetrating Lasers	The key Rebel weapons tech: every Rebel ship's laser fire now pierces — full damage to the primary target, then reduced damage to up to 3 additional ships behind it in a line. Stopped instantly by any shield (Shroud's Shield Field or the Human Shield Matrix tech) — the beam simply stops there instead of passing through.

Shared: a Research Center-type building/tech that lets a body research two projects at once instead of one.

Research levels (ships and buildings)

Most ships and buildings aren't a single unlock-or-don't — they upgrade through numbered **levels** via research, each level taking more turns than the last:

- **Level 1** — the base stats/effects already described elsewhere in this document. No research needed; this is what you have by default once the ship/building exists.
- **Level 2** — a meaningful upgrade over Level 1: either a stat boost (a proposed baseline: **+25%** to the unit/building's primary stat or output) or, for the ships that have one, the point where their specialist upgrade actually unlocks (Medium → Jumper works as a "Level 2" outcome rather than a separate one-off unlock). Proposed cost: **2 turns**.
- **Level 3** — a further upgrade on top of Level 2 (another **+25%**, or more output from an ability already unlocked). Proposed cost: **3–4 turns** (deliberately pricier than Level 2, since it's the deeper investment).

Two specialists already have their level-scaling fully worked out (Section 11), and make good concrete templates for tuning everything else:

- **Shroud's Shield Field** — Level 1: 5 shield HP / 3-field radius. Level 2: 7 shield HP / 4-field radius. Level 3: 9 shield HP / 5-field radius.
- **Jumper's Phase Jump charge time** — Level 1: ~1 second. Level 2: 0.75 seconds. Level 3: 0.5 seconds (each level makes the ability *faster to use*, rather than bigger/stronger — a reminder that "+25%" isn't the only valid shape for a level-up).

Other specialists (Trebuchet's Rocket Barrage cooldown, Lancer's beam falloff, Drone Ship's drones-per-swarm, Aegis/Scarab's heal radius or output) are good candidates for the same treatment, but don't have confirmed per-level numbers yet. Buildings follow the same pattern — a Level 2 Mineral Mine extracts faster, a Level 2 Spaceport builds ships faster, and so on. Exact per-level numbers and per-ship/building applicability need a full balance pass; the three-tier structure and its turn costs are the starting proposal.

14. Commanders (Admirals)

"Commander" and "Admiral" refer to the same system in this design — every army/fleet, for either faction, can have one assigned.

- Each faction has its own pool of **10 commanders** (Human commanders only lead Human fleets, and vice versa).
- An army/fleet can have **one commander assigned** at a time, chosen from a fleet command panel that shows the commander's portrait and exact bonuses before assignment.
- Commanders grant bonuses to some combination of Attack, HP, Speed, Accuracy, and Rate of Fire — each commander has their own distinct strengths, not a generic bonus package.
- Commanders progress through **1 to 5 stars**, scaling their bonuses — this should only ever change as the result of an actual in-game progression event, never silently.

- In battle, the assigned commander's portrait and name are shown, and commanders have their own authored voice lines/taunts they use toward each other during a fight.

Progression

- Commanders gain **XP** (the natural source is participating in battles) which accumulates toward their star rank.
- A commander's bonus isn't a single flat stat — it can cover **Speed, Cooldown reduction, Defense, Strength (attack), or a combination** of these. Each commander has their own specific mix, which is what makes picking one for a given fleet an actual decision rather than a flat upgrade every commander provides equally.

Planned, not yet decided: commander special abilities

Beyond passive fleet bonuses, commanders may eventually get their own battle-activatable special ability — ideas raised include a nuke, a heal, or something unique per commander — costing **Energy** to trigger mid- battle. This is a future direction, not a locked decision; which commanders get which ability, exact costs, and cooldowns are all still open.

This system needs 20 total commander portraits/identities (10 per faction) at final asset scope.

15. Discoverable content: derelict stations

Scattered through the galaxy are derelict station wrecks the player can investigate. Choosing to act on one costs a modest amount of Minerals (example values) and grants exactly one of:

Choice	Example cost (Minerals)	Result
Salvage	5	A burst of research points
Crew	8	A couple of free light ships (the faction's cheapest combat hull)
Repair	15	Claim the hulk as an owned space station (Section 6)

16. Current implementation status (read before assuming a system exists)

This document describes the **full target design**. As of this writing, the only version of this game that's actually playable is a **real-time** prototype built as a total-conversion mod of Star Ruler 2 — it does **not** yet implement the notebook-doodle art style, the turn structure, the separate tactical battle map, prep phase, army-vs-army proximity triggers, body defenses, commanders, or finite per-body Minerals described above. In that current build:

- The visual style is Star Ruler 2's own 3D sci-fi rendering, not hand-drawn doodles — none of Section 2's art direction is implemented yet.
- Everything happens continuously in real time, not in discrete turns.
- Combat is Star Ruler 2's native continuous space combat — there is no separate battle map, no prep phase, and no manual per-unit deployment screen.
- The map is Star Ruler 2's native free-form starfield, not a square grid.
- Minerals and Energy are both produced indefinitely by buildings once built (no finite per-body pool yet, no Energy upkeep yet).
- Conquest is a single channeled "Conquer" action (8 real-time seconds) instead of a turn-based combat-then-claim sequence, and enemy-owned bodies must have their buildings destroyed first via a separate "Bombard" action.
- Body defenses (Rocket Battery/Laser/Forcefield/Disruptor), the 50-ship army cap, commanders, all ship R-abilities, the Capitol's strategic Jump, stargates, battle-wide superweapons, and per-body finite Minerals are not implemented. The current build's ship roster is also different and faction-symmetric (both factions share a Small/Medium/Large lineup plus one faction-exclusive support hull), unlike the asymmetric Human vs. Rebel roster described in Section 11.

None of that invalidates this document — it's the direction the full game is meant to go — but **assets should be built for the design described in Sections 1–15**, and anyone extending the current codebase should treat the gaps above as the backlog, not as evidence the design changed.

See `PROJECT_NOTES.md` (same folder) for the engineering-side detail on that current build.

17. Visual & asset reference

Reference images should be attached here by category, once provided. See Section 2 first for the overall art direction (notebook doodle style, flip-book animation, drawn shadows/comic lines/stick-figure ejection, and the Human blue-metal vs. Rebel rusty-earthly color split) — everything below should be built to match that, not as a separate style.

Faction identity

- **Humans** — metal, blue, sophisticated/strong (Section 2).
- **Rebels** — rusty, patched-up scrapyard, earthy tones (Section 2).

(Add faction emblem/logo doodle images here once provided.)

The galaxy map

- A grey/blank doodle treatment for undiscovered **areas**, and a fully revealed square-grid "notebook paper" look once an area is entered — at least a few doodled variants (empty space, nebula/asteroid field, near a star) for the 10×10 grid inside an area.

- Per-body doodle icons: planet (3×3), moon (2×2), asteroid belt (3×3), star, station, black hole — each drawn, not rendered, sized to the field footprints in Section 3.
- Stargate structure icon/model (strategic map).

Battle environments

Since each tactical battle's map reflects the strategic tile it happened on, at minimum: open space, near-a-star, near-an-asteroid-field, and near-a-planet/station battlefield backdrops — all on the same square-grid notebook-paper look as the strategic map (Section 10).

Propulsion, weapons & VFX identity

The two factions should read as different at a glance from engine trails and weapons fire alone, before a player even sees the hull shape:

- **Humans — combustion/fire-based.** Engines are rough, wild-fire-style burning exhaust, not clean energy trails. Damage/destruction effects lean into fire: sharp radiating explosion bursts plus billowing, scribbly smoke trails on a dying ship — plus the ejected-stick-figure gag from Section 2 on destruction.
- **Rebels — ion-thruster/laser-based.** Propulsion reads as a cleaner, more energetic ion-drive trail (rather than combustion), and their weapons are laser bolts (parallel twin-line beam shots) rather than projectile fire. Impact/destruction effects lean into sparks and debris at the hit point rather than fire and smoke.

This split should carry through every Human vs. Rebel ship, not just the big flagships — it's as much a propulsion/engine-FX identity as it is a hull-design one.

Ships

Each needs a portrait/icon and an exterior model/silhouette concept. Note the rosters are **not mirrored** — see Section 11. Concept sketches exist for the full roster (see `concept_art/` in this folder — add the source image there under a descriptive filename, e.g. `concept_art/ship_roster_sketch_01.jpg`, so it can be referenced directly alongside this text) and describe the following silhouettes and behavior:

Human roster (from concept sketch):

- **Small** — a simple small oval/rounded hull; plain straight engine-trail dash beneath it, no flourish (baseline ships keep their trail simple).
- **Medium** — a small hook/"C"-shaped hull, open bracket profile.
- **Large** — a blockier "E"-shaped hull: flatter top, a notch cut into the profile, visibly bigger than Small/Medium.
- **Shroud** — a diamond/kite-shaped hull with two triangular points, like a stylized gem or emblem — reads as a shield/utility unit rather than a gun platform. Its sketched effect is a jagged, spiky radiant burst beneath it (distinct from the plain baseline trail), tying its visual identity to its shield ability rather than to thrust.

- **Artillery** (heavy unit) — a turreted platform: a rectangular base with a raised box/turret element on top. Its R-ability (homing rockets) is sketched as a separate small rocket shape with its own short exhaust trail, distinct from the ship's own engine trail.
- **Aegis** — a small hull with a raised antenna/dish protruding from the top (the source of its healing radius), reusing the existing "Repair Ship" concept sketch as its base silhouette. Sketched with a dotted-line connection reaching toward another small ship shape — the visual for its continuous radius-2 repair effect.
- **Trebuchet** — no concept sketch yet (added after the original roster sketch). Needs a heavy, slow-looking siege silhouette distinct from Artillery's turreted platform — think an oversized, top-heavy launcher arm rather than a boxy turret, to read as "even slower and heavier than Artillery" at a glance.
- **"S-ship upgraded"** — the roster sketch separately labels one small-hull icon this way, distinct from the plain baseline Small entry. It isn't clear yet whether this means a visual-only upgraded look for the baseline Small hull, or an early alternate name/idea for one of the specialists above (most likely Shroud, since it's drawn nearby). **Flagged for confirmation** rather than guessed at here.
- Fire/smoke reference sketches: a sharp radiating explosion burst, and a separate scribbly "smoke" cloud — the two damage/destruction effects to build out for Human ships.

Rebel roster (from concept sketch):

- **Medium** — a flat, low rectangular hull with small fin-like protrusions; sleeker and flatter than the Human Medium.
- **Capitol Ship** — a large hull made of three joined rectangular segments forming one elongated shape, dramatically bigger than any other Rebel ship — reads immediately as "carrier-scale," consistent with its role as the Rebels' mobile home base.
- **Jumper** — a small hull with a wing/tail fin, sketched with a directional arrow beside it emphasizing speed/movement — needs a visual "upgraded" tell versus the plain Medium once Phase Jump is researched (glowing drive/phase emitters work thematically).
- **Drone Ship** — an elongated hull with a jagged ridge/spine of small fins along the top, like a launcher array — reads as a carrier/launcher for its drones.
- **Drones** (the ones Drone Swarm deploys) — sketched as a plain, minimal small square: deliberately generic and simple, since there are meant to be several on screen at once.
- **Scarab** — reuses the existing "H"-shaped small-hull concept sketch (two prongs joined by a bar), previously assigned to the retired Suicide Drone concept — visually distinct from the plain-square swarm drones, and fast/agile-reading to match its high Speed 3.5 stat.
- **Scarab Repair Drone** — a tiny, minimal drone shape, visually similar to a Microdrone but with a distinct healing-beam/cross-mark detail so players don't confuse "healing swarm" with "attacking swarm" at a glance.
- **Lancer** — no concept sketch yet (added after the original roster sketch, alongside Trebuchet). Needs an extreme-range beam-siege silhouette that reads as the Rebel counterpart to Trebuchet — sleeker and more angular than Trebuchet's heavy launcher-arm look, consistent with the faction's ion/laser identity, but similarly slow-moving.
- Laser/impact reference sketches: a laser bolt (parallel twin-line beam with a directional arrow) for weapons fire, and a separate impact sketch showing a small ship struck with sparks/debris radiating outward — the damage/destruction effect to build out for Rebel ships (sparks/debris, not

fire/smoke).

(Add ship portrait/model images here once provided.)

Buildings and defenses

- Economic: Mineral Mine, Energy Harvester (basic), Specialist Asteroid Harvester, Specialist Stellar Harvester, Spaceport, Research Lab.
- Defense platforms (visually distinct category from economic buildings): Rocket Battery, Laser, Forcefield (needs a shield-bubble visual effect), Disruptor (needs a periodic pulse/jam visual effect).

(Add building/defense icon images here once provided.)

Resource icons

- **Minerals** — a gem/ore icon (not a coin, not a hammer/tool icon).
- **Energy** — an energy/bolt-style icon.

Commanders

- 10 Human commander portraits, 10 Rebel commander portraits, plus a visual language for the 1–5 star progression.

Battle-wide ability icons

- Antimatter Bomb / Thermonuclear Device activation icon and its large-area detonation visual effect.

UI elements

- Empty/filled build-slot icons, and a visually distinct empty/filled defense-slot icon set (so players don't confuse the two slot types).
- Ship-build button icon.
- Top-left resource readout: current Minerals/Energy plus a small "+" next-round-projection indicator for each.
- Bottom-right fleet panel: a row of per-ship slots (reorderable by strength/defense) plus one dedicated Commander slot.
- A "Claim" action button/icon, plus a way to visually show a body's buildings being ground down (defense value depleting) when claiming an enemy body that has no defense platform.
- Prep-phase deployment screen chrome (ready button, per-side deployment zone coloring).
- In-battle unit portraits (shown per ship, disappear as ships die).
- R-ability icon per specialist ship (Shroud, Artillery, Trebuchet, Jumper, Drone Ship, Lancer) plus cooldown-timer treatment. Aegis and Scarab have no manual R-ability (Aegis is fully automatic; Scarab's drone deploy is reactive, not player-triggered), so they need a passive "active/healing" status indicator instead of a cooldown icon.

- Rocket-interception visual: a distinct small "shot down" spark/burst effect for when a Shroud shield, Microdrone, or Scarab Repair Drone intercepts an incoming rocket from Trebuchet's Rocket Barrage.

(Add all image references here once provided.)

Where this fits alongside other docs

- `PROJECT_NOTES.md` (same folder) is the **engineering** handoff for the current real-time Star Ruler 2 prototype specifically — code structure, testing protocol, gotchas.
- This file is the **full design/asset** reference — build assets against this document, not against whatever subset happens to be implemented in any one codebase at a given time.
- The canonical written source for the mechanics in this document is `GAME_DESIGN_SOURCE_OF_TRUTH.md` in the `BergerBerger/planet-game` GitHub repository. Where this file and that one differ, the newest explicit design instruction wins, per that document's own stated rule.